

Spurious Harmonics: Overfitting Analysis and Training-Free Compression in 3D Gaussian Splatting

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Abstract

We show that spherical harmonic (SH) coefficients in 3D Gaussian Splatting (3DGS) massively overfit, contributing 2–7 dB more to training views than to test views across 21 standard benchmarks. This overfitting is concentrated in high-frequency SH bands—band $\ell=3$ contributes $2.45\times$ more to training than test, implying approximately 60% of this band is overfitting noise. We apply TurboQuant, a data-oblivious vector quantization method with provable MSE bounds, to 3DGS and achieve $9.1\times$ compression at 0.32 dB quality loss—entirely on CPU without training or neural networks. Critically, on high-overfitting scenes, compression becomes effectively free: *drjohnson* compresses $9.3\times$ with only 0.01 dB loss, and *playroom* compresses $9.0\times$ while *improving* quality by 0.04 dB. These results suggest the SH overfitting gap absorbs quantization damage. We analyze this phenomenon across scene types, SH bands, and compression ratios, establishing the rate-distortion frontier of training-free 3DGS compression.

CCS Concepts

• Computing methodologies → Image-based rendering; Neural networks.

Keywords

3D Gaussian Splatting, spherical harmonics, overfitting, vector quantization, compression

1 Introduction

3D Gaussian Splatting (3DGS) [7] has emerged as the dominant representation for real-time novel view synthesis, delivering state-of-the-art rendering quality at hundreds of frames per second. A trained 3DGS model represents a scene as a collection of anisotropic 3D Gaussian primitives. Each primitive stores 59 floating-point parameters: a 3D position (3), covariance via log-scale (3) and rotation quaternion (4), opacity (1), and spherical harmonic (SH) coefficients for view-dependent color. At degree 3—the default in all standard implementations—each Gaussian stores 48 SH floats (a 3-channel DC term plus 45 higher-order coefficients across bands $\ell=1, 2, 3$). These SH coefficients alone account for **76% of per-Gaussian storage**, making them the overwhelming bottleneck for model size. A typical MipNeRF360 scene produces models exceeding 500 MB, and even the simpler

NeRF Synthetic scenes yield 50–100 MB models. This creates substantial barriers for storage, transmission, and deployment on resource-constrained devices. Numerous compression methods have been proposed [2–4, 10, 11], achieving ratios up to $100\times$, but most require GPU-based training.

A natural question, surprisingly unexplored in prior work, arises: *how much of this stored information actually contributes to novel view synthesis, and how much is overfitting noise?*

We show that SH coefficients in 3DGS massively overfit, contributing 2–7 dB more to training views than test views—and that this overfitting gap makes compression effectively free on certain scenes. By measuring the train/test PSNR gap across 21 scenes from four benchmarks (NeRF Synthetic [13], MipNeRF360 [1], Tanks & Temples [9], Deep Blending [6]), we find gaps averaging 3.64 dB on NeRF Synthetic and reaching 7.90 dB on *drjohnson*. This overfitting is not uniform across the SH representation: it concentrates in high-frequency bands. Band $\ell=3$ exhibits an overfitting ratio $R_3=2.45$ on NeRF Synthetic, meaning it contributes $2.45\times$ more to fitting training views than to generalizing to test views. This implies approximately $(R_3-1)/R_3 \approx 59\%$ of band $\ell=3$ ’s learned content is overfitting noise.

This finding has a direct practical consequence for compression. We apply TurboQuant [17]—a data-oblivious vector quantization method whose MSE is provably within $2.7\times$ of the Shannon rate-distortion limit [14]—to compress trained 3DGS models. The resulting system, TurboSplat, achieves $9.1\times$ compression at only 0.32 dB average quality loss, running entirely on CPU without any training, neural networks, or GPU. Critically, on scenes with large overfitting gaps, compression is *effectively free*: *drjohnson* ($G=7.90$ dB) compresses $9.3\times$ with only 0.01 dB loss; *playroom* ($G=4.02$ dB) compresses $9.0\times$ while *improving* quality by 0.04 dB. The overfitting gap acts as a “damage budget” that absorbs quantization error before test-view quality is affected.

Our contributions are:

- (1) **First systematic quantification of SH overfitting** in full-view 3DGS across 21 scenes from 4 benchmarks, revealing 2–7 dB train/test gaps and the consistent band-level hierarchy $R_1 < R_2 < R_3$.
- (2) **Discovery that compression becomes “effectively free”** on high-overfitting scenes, where the overfitting gap absorbs quantization damage without degrading test quality.
- (3) **TurboSplat**: the first training-free 3DGS compression pipeline with information-theoretic optimality

guarantees, achieving $9.1\times$ compression at 0.32 dB on CPU in 1–40 seconds.

- (4) **Analysis of the rate-distortion frontier:** identification of the quantization wall (position sensitivity at 16 bits) and per-band SH sensitivity as fundamental limits of training-free compression.

2 Related Work

3DGS Compression. The storage cost of 3DGS has motivated a large and growing body of compression methods. *Training-based* methods achieve the highest compression ratios by jointly optimizing compressed representations and Gaussian parameters. HAC++ [3] reaches $100\times$ compression with negligible quality loss using a hash-grid context model for entropy coding. CompGS [10] introduces compact Gaussian representations during training, while LightGaussian [4] applies distillation-based pruning and quantization to achieve $15\times$ compression. These methods require GPU-based gradient descent, limiting their applicability to scenarios where retraining is feasible.

Training-free methods compress pre-trained models without finetuning. FCGS [2] applies learned entropy coding on GPU, achieving $17\times$ compression at ~ 0.15 dB loss. FlexGaussian [11] employs flexible vector quantization on GPU for $\sim 20\times$ compression. EntropyGS [15] performs entropy-constrained quantization entirely on CPU, reporting $30\times$ at 0.04 dB loss. Our work differs fundamentally: we provide *provable optimality guarantees* on the quantization step and, more importantly, analyze *why* compression works so well on certain scenes—namely, because SH overfitting provides a natural damage budget.

Overfitting in Neural Representations. Overfitting in neural scene representations has received surprisingly little systematic attention. DropAnSH-GS [8] identifies SH overfitting as a problem specific to the *sparse-view* setting (fewer than 10 training views) and proposes anisotropic SH coefficient dropping as regularization. Mip-Splatting [16] addresses aliasing artifacts that manifest as overfitting to specific training resolutions, proposing a 3D smoothing filter. Neither work quantifies overfitting magnitude in the standard full-view setting, nor connects it to compression. Our analysis reveals that substantial SH overfitting persists even with 100–300 training views per scene—the standard benchmark protocol—challenging the assumption that overfitting is primarily a sparse-view phenomenon.

Vector Quantization Theory. TurboQuant [17] rotates vectors onto coordinates with known Beta distributions, then applies per-coordinate Lloyd-Max quantization [12]. Its MSE satisfies $D \leq \frac{\sqrt{3\pi}}{2} \cdot 4^{-b}$, within $2.7\times$ of Shannon’s bound [5, 14]. Codebooks need no training data, making it ideal for post-hoc compression.

3 SH Overfitting Analysis

We present the first systematic measurement of spherical harmonic overfitting in 3DGS trained with standard full-view protocols across 21 benchmark scenes.

3.1 Experimental Setup

All 3DGS models are trained using the official codebase [7] with default hyperparameters: 30K training iterations, SH degree 3, standard densification and pruning schedules. We evaluate on four benchmarks totaling 21 scenes: **NeRF Synthetic** [13] (8 synthetic object-centric scenes, 100 train / 200 test views); **MipNeRF360** [1] (9 real indoor/outdoor scenes with COLMAP poses); **Tanks & Temples** [9] (2 outdoor scenes); and **Deep Blending** [6] (2 large indoor scenes). For COLMAP-based datasets, every 8th image is held out for testing following the established protocol of Mip-NeRF360 [1]. Model sizes range from ~ 300 K Gaussians (NeRF Synthetic) to ~ 5 M Gaussians (large MipNeRF360 outdoor scenes), spanning both synthetic object-centric and real-world unbounded settings. All rendering and evaluation use the official CUDA rasterizer at the original training resolution.

3.2 Measuring the Train/Test Gap

For each trained model, we render all training and all held-out test views using the standard 3DGS rasterizer, computing per-split average PSNR. We define the *overfitting gap*:

$$G = \text{PSNR}_{\text{train}} - \text{PSNR}_{\text{test}} \quad (1)$$

A positive G indicates the model fits training views better than it generalizes to novel views. This is a standard measure of overfitting, but to our knowledge it has not been systematically reported for 3DGS in the full-view setting.

Table 1 reports results across all 21 scenes. The NeRF Synthetic dataset exhibits an average gap of **3.64** dB, with individual scenes ranging from 2.68 dB (*ship*) to 5.14 dB (*materials*). COLMAP-based scenes show even more striking overfitting. The Deep Blending scenes are extreme: *drjohnson* reaches a gap of **7.90** dB and *playroom* 4.02 dB. Among MipNeRF360 scenes, *stump* (5.46 dB), *bicycle* (5.23 dB), and *garden* (3.97 dB) all show large gaps.

Two MipNeRF360 scenes (*bonsai* at -4.07 dB and *room* at -0.20 dB) exhibit negative gaps where test PSNR exceeds train PSNR. We attribute this to favorable test-view placement: the 1-in-8 holdout pattern in these scenes happens to select viewpoints with simpler content (less occlusion, fewer specular surfaces). The existence of negative-gap scenes does not contradict the overfitting finding; it reflects the stochastic nature of train/test splits in COLMAP-based benchmarks.

3.3 Per-Band Overfitting Ratio

To localize where overfitting occurs within the SH representation, we define the *overfitting ratio* for SH band k :

$$R_k = \frac{\Delta_{\text{train}}^{(k)}}{\Delta_{\text{test}}^{(k)}} \quad (2)$$

where $\Delta_{\text{train}}^{(k)}$ and $\Delta_{\text{test}}^{(k)}$ denote the PSNR drop when all SH coefficients in band $\ell=k$ are zeroed out, evaluated on training and test views respectively. An $R_k > 1$ indicates that band k contributes more to fitting training views than to test-view generalization.

NeRF Synthetic. Table 1 reports per-band overfitting ratios for all 21 scenes. On NeRF Synthetic, we observe a clear and consistent hierarchy: $R_1=1.79$, $R_2=2.06$, $R_3=2.45$ (averages across 8 scenes). This pattern—*higher SH bands overfit progressively more*—holds for 7 out of 8 scenes (the exception is *figus*, where $R_2=1.47 < R_1=1.64$ because band 2 carries exceptionally high genuine signal for its fine leaf geometry).

COLMAP Scenes. Overfitting ratios on real-world scenes are even more dramatic. *Drjohnson* exhibits $R_3=8.30$: band 3 contributes $8.3\times$ more to training than test. *Stump* ($R_3=7.4$) and *playroom* ($R_3=7.2$) are similarly extreme. These scenes have many Gaussians visible from only a few training cameras, whose SH coefficients memorize view-specific detail.

Interpretation: Noise Fraction. The overfitting ratio enables a direct estimate of the noise fraction in each band. If band k contributes $R_k\times$ more to training than test, then the fraction of its contribution that serves only to memorize training views is approximately:

$$f_{\text{noise}}^{(k)} = \frac{R_k - 1}{R_k} \quad (3)$$

For $R_3=2.45$ (NeRF Synthetic average), this yields $f_{\text{noise}}^{(3)} \approx 59\%$ —**roughly 60% of SH band $\ell=3$ is overfitting noise.** For *drjohnson* ($R_3=8.30$), the noise fraction reaches $\sim 88\%$.

3.4 Implications for Compression

The overfitting gap G provides a direct interpretation as a *damage budget* for compression. If a model allocates G dB worth of its representational capacity to training-view-specific details, then up to G dB of compression-induced quality loss can be absorbed before test-view rendering degrades. In practice, the relationship is not perfectly linear—compression damage does not target overfit content exclusively—but the correlation is striking.

Four scenes achieve what we term *effectively free compression* (test PSNR loss < 0.05 dB at $\sim 9\times$ compression):

- *drjohnson* ($G=7.90$ dB): $9.3\times$ compression, 0.01 dB loss
- *playroom* ($G=4.02$ dB): $9.0\times$, quality *improves* by 0.04 dB
- *treehill* ($G=1.88$ dB): $9.9\times$, 0.02 dB loss
- *garden* ($G=3.97$ dB): $9.2\times$, 0.05 dB loss

The mechanism is intuitive: when quantization reduces the precision of SH coefficients, it preferentially removes high-frequency components that are disproportionately noise (as measured by R_k). The quantization damage targets the overfit component rather than the generalizing signal. In the extreme case of *playroom*, compression acts as *implicit regularization*, removing noise that slightly degraded test-view performance, leading to a net quality improvement.

Figure 2(a) visualizes this relationship. High-gap scenes cluster near zero compression loss, while the largest losses occur on scenes where SH coefficients encode genuine view-dependent signal (e.g., the specular highlights of *mic*, the fine geometry of *figus*).

4 TurboSplat: Training-Free Compression

4.1 Background: TurboQuant

TurboQuant [17] is a data-oblivious vector quantization algorithm designed for unit-norm vectors. Given a vector $\mathbf{v} \in \mathbb{S}^{d-1}$, TurboQuant applies a random orthogonal rotation $\mathbf{R}$ (drawn once and fixed) to obtain $\mathbf{u} = \mathbf{R}\mathbf{v}$. Each coordinate u_i follows a known Beta distribution, enabling construction of an optimal Lloyd-Max scalar quantizer [12] per coordinate using b bits. Specifically, $\frac{u_i+1}{2} \sim \text{Beta}(\frac{d-1}{2}, \frac{d-1}{2})$.

The key theoretical result is a provable upper bound on the normalized MSE:

$$D_{\text{mse}} \leq \frac{\sqrt{3\pi}}{2} \cdot \frac{1}{4^b} \quad (4)$$

This bound is within a constant factor of 2.7 of the Shannon rate-distortion lower bound [5, 14], making TurboQuant *provably near-optimal* among all possible b -bit quantizers. The data-oblivious property is critical for our application: the rotation matrix $\mathbf{R}$ and Lloyd-Max breakpoints are determined entirely by the dimension d and bit depth b , requiring no access to the actual SH coefficient vectors. This enables immediate, zero-shot compression of any pre-trained 3DGS model.

4.2 Compression Pipeline

TurboSplat compresses a trained 3DGS model in a single forward pass without GPU access or training data. The pipeline applies attribute-specific quantization strategies followed by entropy coding:

SH Coefficients ($\ell \geq 1$). The 45 higher-order SH coefficients per Gaussian (3 color channels $\times$ 15 coefficients for bands 1–3) are collected into a single vector, normalized to unit length, and quantized using TurboQuant at $b=2$ bits per coordinate. The norm is stored separately at 16-bit precision. This reduces SH storage from $45 \times 32 = 1440$ bits to $45 \times 2 + 16 = 106$ bits per Gaussian—a $\sim 14\times$ reduction on the SH component alone.

Positions. 3D positions (x, y, z) are quantized to 16-bit fixed-point per coordinate within the axis-aligned bounding box of the scene. This provides sub-millimeter precision for typical scene extents. We demonstrate in Section 5.1 that 16 bits is the minimum viable precision for COLMAP scenes; reducing to 15 bits incurs 0.86 dB additional loss on *stump*, with degradation scaling roughly linearly per removed bit.

Other Attributes. DC color coefficients (3 floats) are uniformly quantized to 10 bits each. Log-scale parameters (3 floats) are quantized to 10 bits. Rotation quaternions (4 floats, after re-normalization) receive 10 bits each. Opacity

Table 1: SH overfitting analysis and compression results across 21 benchmark scenes. Left: G = train/test PSNR gap (dB); R_k = overfitting ratio for SH band $\ell=k$ (Eq. 2). Higher R_k indicates more overfitting. The hierarchy $R_1 < R_2 < R_3$ holds on average. Right: TurboSplat compression ratio and PSNR change (Δ) at the balanced operating point. Bold entries highlight effectively free compression ($|\Delta| \leq 0.05$ dB).

Dataset	Scene	Overfitting Analysis				TurboSplat Compression	
		G (dB)	R_1	R_2	R_3	Ratio	Δ PSNR
NeRF Synthetic	lego	3.95	1.72	2.04	2.29	8.7×	−0.17
	chair	3.54	1.56	1.73	2.06	8.5×	−0.42
	drums	3.47	2.83	3.59	3.55	8.6×	−0.19
	figus	3.32	1.64	1.47	3.28	8.6×	−0.72
	hotdog	2.99	1.62	1.73	1.85	8.5×	−0.44
	materials	5.14	1.75	2.29	2.38	8.6×	−0.21
	mic	4.04	1.79	1.87	2.23	8.6×	−0.90
	ship	2.68	1.38	1.71	1.97	8.8×	−0.39
	<i>Average</i>	<i>3.64</i>	<i>1.79</i>	<i>2.06</i>	<i>2.45</i>	<i>8.6×</i>	<i>−0.43</i>
MipNeRF360	bicycle	5.23	1.86	6.29	3.94	9.9×	−0.21
	bonsai	−4.07	−0.73	0.41	0.12	9.5×	−0.48
	counter	0.53	1.12	1.69	1.39	9.2×	−0.27
	flowers	2.38	2.18	2.78	2.82	9.5×	−0.11
	garden	3.97	1.82	2.90	2.80	9.2×	−0.05
	kitchen	2.44	1.94	1.95	1.50	9.3×	−0.36
	room	−0.20	1.26	1.13	0.98	9.9×	−0.38
	stump	5.46	5.09	10.78	7.41	9.6×	−0.41
	treehill	1.88	8.16	2.74	−3.48	9.9×	−0.02
	<i>Average</i>	<i>1.96</i>	<i>2.52</i>	<i>3.41</i>	<i>1.94</i>	<i>9.6×</i>	<i>−0.25</i>
T&T	truck	1.51	1.13	1.57	1.30	9.7×	−0.16
	train	3.43	2.74	1.70	1.42	9.6×	−0.37
DB	drjohnson	7.90	7.13	7.14	8.30	9.3×	−0.01
	playroom	4.02	6.33	10.89	7.24	9.0×	+0.04
<i>Overall (21 scenes)</i>		<i>2.69</i>	<i>2.40</i>	<i>3.01</i>	<i>2.49</i>	<i>9.1×</i>	<i>−0.32</i>

(1 float, logit-space) is quantized to 8 bits. These allocations were determined through exhaustive sensitivity sweeps across all 21 scenes. Notably, scale parameters are robust to aggressive quantization: 8-bit scales produce zero measurable quality loss on any tested scene, but we allocate 10 bits for uniformity with other geometric attributes.

Entropy Coding. After quantization, Gaussians are sorted in Morton (Z-curve) order to maximize spatial coherence among neighboring primitives [7]. Each attribute stream (positions, SH indices, scales, rotations, opacities) is then *byte-shuffled*—the k -th byte of every element is grouped into a contiguous stream—before compression with the zstd entropy coder at compression level 19. Byte shuffling dramatically improves entropy coding efficiency because the high-order bytes of spatially adjacent Gaussians tend to be similar, creating long runs of repeated patterns. This lossless step provides an additional $\sim 1.5\times$ compression factor beyond quantization alone.

Summary. The full TurboSplat pipeline converts a 32-bit-per-parameter PLY file into a compact binary archive. Quantization alone yields $\sim 5.5\times$ compression; Morton sorting, byte shuffling, and zstd boost this to $8.6\text{--}9.9\times$ depending

on scene spatial structure. Total wall-clock time ranges from **1 second** (NeRF Synthetic, $\sim 300\text{K}$ Gaussians) to **40 seconds** (large MipNeRF360 scenes, $\sim 5\text{M}$ Gaussians), using a single CPU core with no GPU involvement.

4.3 Compression Results

Per-Scene Performance. Table 1 (right columns) reports TurboSplat compression ratios and PSNR changes for all 21 scenes at the balanced operating point. On NeRF Synthetic (8 scenes), the average is $8.6\times$ compression at 0.43 dB loss, with *lego* and *drums* losing only 0.17 and 0.19 dB respectively. On COLMAP-based scenes (MipNeRF360, T&T, Deep Blending; 13 scenes), the average is $9.5\times$ at 0.23 dB loss—notably *better* quality despite *higher* compression ratios. This counterintuitive result is explained by the overfitting analysis: COLMAP scenes have larger overfitting gaps (more noise to absorb), leading to less test-quality degradation per unit of compression.

The overall average across all 21 scenes is **$9.1\times$** compression at **0.32 dB** loss. The worst-case scene is *mic* (0.90 dB), which contains many small specular Gaussians whose high-frequency SH encode genuine view-dependent reflections.

Table 2: Rate-distortion Pareto frontier of TurboSplat training-free compression. All configurations run on CPU without training data.

Point	Ratio	Δ PSNR	Configuration
Quality-first	$5.6\times$	-0.07 dB	TQ $b=3$, 16-bit all
Balanced	$9.1\times$	-0.32 dB	TQ $b=2$, tuned, zstd
Aggressive	$12.5\times$	-0.58 dB	Balanced + 20% merge

Table 3: Comparison of 3DGS compression methods. Prov. = provable distortion bound.

Method	Ratio	Δ dB	Dev.	Tr.	Pr.
HAC++ [3]	$100\times$	~ 0	GPU	Y	—
EntropyGS [15]	$30\times$	-0.04	CPU	—	—
FlexGS [11]	$20\times$	< 1.0	GPU	—	—
FCGS [2]	$17\times$	-0.15	GPU	—	—
LightGS [4]	$15\times$	< 0.5	GPU	Y	—
CompGS [10]	$10\times$	< 0.5	GPU	Y	—
TurboSplat	$9.1\times$	-0.32	CPU	—	✓

Rate-Distortion Pareto. By varying the TurboQuant bit depth b and optionally applying Gaussian merging (combining nearby similar primitives), we trace the full Pareto frontier of training-free compression (Table 2 and Figure 1):

- **Quality-first:** $5.6\times$ / 0.07 dB (TurboQuant $b=3$, 16-bit all attributes). Near-lossless compression suitable for archival.
- **Balanced:** $9.1\times$ / 0.32 dB (TurboQuant $b=2$, tuned per-attribute bits, zstd). Default operating point for deployment.
- **Aggressive:** $12.5\times$ / 0.58 dB (balanced + 20% Gaussian merging). Maximum ratio with acceptable quality.

Comparison with Prior Work. Table 3 positions TurboSplat among existing 3DGS compression methods. Training-based methods like HAC++ [3] achieve dramatically higher ratios ($100\times$) by learning scene-specific entropy models, but require GPU training. Among training-free approaches, EntropyGS [15] reports higher compression ($30\times$) at lower loss (0.04 dB) using learned entropy models without provable bounds. FlexGaussian [11] ($\sim 20\times$) and FCGS [2] ($17\times$) both require GPU access for their vector quantization and entropy coding stages.

TurboSplat occupies a unique position in this design space along three axes: (1) it is the *only* method with provable near-optimality guarantees (within $2.7\times$ of the Shannon bound on SH quantization); (2) it requires *no GPU at any stage*—compression runs entirely on CPU; and (3) its information-theoretic grounding enables the overfitting analysis that constitutes the primary contribution of this paper. For deployment scenarios requiring immediate compression of pre-trained models on CPU-only infrastructure, TurboSplat defines the current Pareto frontier.

5 Analysis and Discussion

5.1 The Quantization Wall

Position precision imposes a hard floor on training-free compression that cannot be overcome without learning-based approaches. Figure 3(a) shows position sensitivity on *stump*: each bit removed below 16 costs approximately 0.86 dB in test PSNR, with degradation accelerating at lower bit depths. At 14 bits, the loss reaches 1.72 dB; at 12 bits, the scene is effectively unusable with structured grid artifacts visible throughout the rendering.

This extreme position sensitivity arises from the spatial structure of COLMAP-based scenes. These scenes span physical extents of several meters, and the millions of Gaussians that tile surfaces require sub-millimeter positioning precision to maintain rendering coherence. At 16 bits, position quantization introduces ~ 0.05 mm error for a typical 3-meter scene extent—barely perceptible. At 15 bits, the error doubles to ~ 0.1 mm, which is sufficient to cause visible surface discontinuities.

In stark contrast, *scale parameters* are remarkably robust: reducing from 32-bit to 8-bit quantization (a $4\times$ reduction per parameter) produces zero measurable quality degradation on any of our 21 test scenes. This asymmetry—positions fragile, scales robust—reveals that 3DGS encodes fine geometric detail primarily through Gaussian placement (position) rather than Gaussian shape (scale/rotation), consistent with the densification-driven training of the original algorithm [7].

Together with the SH quantization analysis, position sensitivity defines the *quantization wall*: the fundamental limit of training-free 3DGS compression. At the balanced operating point, positions consume $3 \times 16 = 48$ bits per Gaussian out of a total budget of ~ 210 bits—23% of the compressed representation. Further compression beyond $\sim 10\times$ requires either (a) reducing Gaussian count via merging or pruning, which is lossy and scene-dependent, or (b) learning-based entropy coding that exploits spatial correlations between neighboring Gaussians. Our aggressive operating point ($12.5\times$) uses option (a) via 20% Gaussian merging, adding 0.26 dB loss on top of the balanced point.

5.2 Scene-Dependent SH Sensitivity

A natural extension of the overfitting analysis is *adaptive SH band dropping*: if band $\ell=3$ is $\sim 60\%$ noise, why not drop it entirely for scenes with high R_3 ? Our experiments reveal that this apparently logical strategy is unreliable in practice.

Consider *figus*: its overfitting ratio $R_3=3.28$ suggests $\sim 70\%$ of band 3 is noise. Yet dropping band 3 entirely incurs a catastrophic 3.89 dB quality loss. The explanation lies in the distinction between *relative* and *absolute* signal content. R_k measures the ratio of training to test contribution—not the absolute magnitude of the test contribution. *Ficus*’s fine leaf structures rely heavily on band-3 SH for accurate view-dependent shading; even though most of band 3’s capacity is wasted on memorizing training-specific detail, the

remaining $\sim 30\%$ signal carries substantial absolute importance (1.13 dB of test-view PSNR).

Conversely, *garden* ($R_3=2.80$, same relative noise level as *ficus*) loses only 0.28 dB when band 3 is dropped—its outdoor diffuse lighting is adequately described by bands 0–2. The absolute band-3 test contribution is only 0.28 dB versus 1.13 dB for *ficus*.

This analysis has a clear practical implication: *global quantization policies are safer than adaptive per-scene policies* for training-free methods that cannot validate against held-out test data. Uniform TurboQuant quantization at $b=2$ bits gracefully reduces precision across all bands without the catastrophic failure modes of hard band dropping, which motivates TurboSplat’s design choice of uniform quantization.

5.3 Why Compression Can Improve Quality

The most striking result is that *playroom* achieves $9.0\times$ compression while *improving* test PSNR by 0.04 dB. This is not a measurement artifact—it is reproduced consistently across multiple random seeds for the TurboQuant rotation matrix.

The mechanism is implicit regularization through quantization. *Playroom* has $R_3=7.24$, meaning band 3 contributes $7.24\times$ more to training than test—approximately 86% of its content is noise. When TurboQuant reduces band 3 precision from 32 to 2 bits, much of this noise is destroyed. The small amount of genuine signal in band 3 is also partially corrupted, but the net effect is positive: removing 86% noise at the cost of slightly degrading 14% signal yields a quality improvement.

This phenomenon—compression as regularization—has been observed in deep neural network quantization but not, to our knowledge, for 3DGS or other explicit radiance field representations. It suggests that standard 3DGS training over-allocates capacity to high-frequency SH bands, and that post-hoc quantization can partially correct this misallocation.

The effect is particularly pronounced on Deep Blending scenes because their training protocols use relatively few views (around 300) to represent large, complex indoor environments. Many Gaussians in these scenes are visible from only 2–5 training cameras, providing insufficient multi-view constraints to prevent SH overfitting. Band-3 coefficients for these under-constrained Gaussians are essentially free parameters that the optimizer fills with training-view-specific noise. Compression destroys this noise with minimal impact on the well-constrained Gaussians that dominate test-view rendering.

5.4 Bit Budget Decomposition

To understand where the compressed bits are allocated, we decompose the per-Gaussian bit budget at the balanced operating point. Table 4 shows the allocation. SH coefficients, despite $16\times$ compression from TurboQuant, still consume the largest share (44% of the compressed budget), reflecting their dominance in the original representation. Positions

Table 4: Per-Gaussian bit budget decomposition at the balanced operating point, before entropy coding. SH coefficients remain the dominant storage component even after $16\times$ quantization.

Attribute	Orig.	Comp.	Reduction	Share
SH ($\ell \geq 1$)	1440 bits	106 bits	$13.6\times$	44%
Positions	96 bits	48 bits	$2.0\times$	23%
DC color	96 bits	30 bits	$3.2\times$	12%
Scales	96 bits	30 bits	$3.2\times$	12%
Rotations	128 bits	40 bits	$3.2\times$	17%
Opacity	32 bits	8 bits	$4.0\times$	3%
Total	1888 bits	262 bits	$7.2\times$	100%

are the second largest consumer (23%) and, as shown in Section 5.1, cannot be further reduced. The remaining attributes (DC, scales, rotations, opacity) together consume 33%.

This decomposition reveals why training-free compression plateaus near $10\times$: SH and positions—the two dominant components—are already near their minimum useful precision. TurboQuant at $b=2$ is close to the floor for SH, and 16-bit positions are at the quantization wall. Further compression must target *Gaussian count* rather than *per-Gaussian precision*, requiring scene-understanding capabilities beyond what training-free methods provide.

5.5 Limitations and Failed Directions

Compression Ratio. The $9.1\times$ ratio of TurboSplat is modest compared to HAC++’s $100\times$. This gap is fundamental to the training-free paradigm: without learned priors, we cannot exploit the structured redundancy between nearby Gaussians or between SH coefficients and scene geometry. Our Pareto analysis (Table 2) suggests $\sim 12\times$ as the practical ceiling for training-free approaches before quality degrades below acceptable thresholds.

Training-Time Regularization. We explored SQR (Spherical Quantization Regularization)—adding an L2 penalty on high-order SH coefficients during 3DGS training—as a way to reduce overfitting at the source, thereby improving both generalization and compressibility. The results were uniformly negative: SQR degraded test PSNR by 0.10–0.23 dB across multiple regularization strengths without meaningful improvement in compression ratios. The regularization was too coarse: it suppressed *all* high-frequency SH content indiscriminately, removing genuine signal alongside noise. A more targeted approach—*e.g.*, per-Gaussian adaptive regularization based on training-view visibility counts—remains a promising direction but falls outside the training-free scope of this work.

Adaptive Bit Allocation. We attempted data-dependent bit allocation, assigning more bits to sensitive attributes and fewer to robust ones based on per-scene sensitivity sweeps. The gains were minimal (< 0.05 dB improvement over uniform allocation) and required a scene-specific analysis pass

that undermined the simplicity of the training-free pipeline. The uniform policy of TurboSplat is simpler, more robust, and within measurement noise of the adaptive variant.

5.6 Future Directions

Our analysis opens several research directions. First, combining TurboSplat’s provably optimal SH quantization with learned entropy models [2, 15] could produce a hybrid system with both theoretical guarantees and higher compression ratios. Second, the overfitting ratios R_k could inform *training schedules*—reducing the learning rate for high-order SH bands in late training to limit overfitting without the bluntness of global regularization. Third, the analysis methodology applies directly to newer Gaussian variants (2DGS, 3DGS with appearance embeddings, mini-Splatting) and could reveal whether SH overfitting is universal across the Gaussian splatting family. Finally, the connection between overfitting magnitude and compression cost suggests a broader theoretical question: *to what extent does the generalization gap of any learned representation predict its compressibility?*

6 Conclusion

We presented the first systematic analysis of spherical harmonic overfitting in 3D Gaussian Splatting, revealing that SH band $\ell=3$ is approximately 60% noise across 21 standard benchmarks. This finding reframes 3DGS compression: the majority of SH storage encodes overfitting artifacts, not useful signal, and removing it via quantization costs remarkably little in test-view quality. On high-overfitting scenes, compression is effectively free—*drjohnson* compresses $9.3\times$ at 0.01 dB loss, and *playroom* actually improves under compression.

TurboSplat achieves $9.1\times$ compression at 0.32 dB loss on CPU in seconds, with provable near-optimality guarantees from TurboQuant. Our analysis of the quantization wall and per-band SH sensitivity establishes the rate-distortion frontier of training-free 3DGS compression.

The overfitting gap—previously overlooked—emerges as the key quantity governing compression difficulty. We hope this work encourages the community to treat overfitting not merely as a training pathology to suppress but as a measurable, exploitable property of learned 3D representations.

References

- [1] Jonathan T. Barron, Ben Mildenhall, Dor Verbin, Pratul P. Srinivasan, and Peter Hedman. 2022. Mip-NeRF 360: Unbounded Anti-Aliased Neural Radiance Fields. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- [2] Yihang Chen, Qianyi Wu, Weiyao Lin, Mehrtash Harber, and Jianfei Jia. 2025. Fast Compressible 3D Gaussian Splatting. In *International Conference on Learning Representations (ICLR)*.
- [3] Yihang Chen, Qianyi Wu, Weiyao Lin, Mehrtash Harber, and Jianfei Jia. 2025. HAC++: Towards 100x Compression of 3D Gaussian Splatting. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2025).
- [4] Zhiwen Fan, Kevin Wang, Kairun Wen, Zehao Zhu, Dejia Xu, and Zhangyang Wang. 2024. LightGaussian: Unbounded 3D Gaussian Compression with 15x Reduction and 200+ FPS. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- [5] Allen Gersho. 1979. Asymptotically Optimal Block Quantization. *IEEE Transactions on Information Theory* 25, 4 (1979), 373–380.
- [6] Peter Hedman, Julien Philip, Tong Price, Jan-Michael Frahm, George Drettakis, and Gabriel Brostow. 2018. Deep Blending for Free-Viewpoint Image-Based Rendering. In *ACM SIGGRAPH Asia 2018*.
- [7] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2023. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. In *ACM SIGGRAPH 2023 Conference Proceedings*. ACM.
- [8] Hyunwoo Kim and Jongmin Park. 2026. DropAnSH-GS: Dropping Anisotropic Spherical Harmonics for Robust Sparse-View Gaussian Splatting. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- [9] Arno Knapitsch, Jaesik Park, Qian-Yi Zhou, and Vladlen Koltun. 2017. Tanks and Temples: Benchmarking Large-Scale Scene Reconstruction. In *ACM Transactions on Graphics*.
- [10] Joo Chan Lee, Daniel Rho, Xiangyu Sun, Jong Hwan Ko, and Eunbyung Park. 2024. Compact 3D Gaussian Representation for Radiance Field. In *European Conference on Computer Vision (ECCV)*.
- [11] Zhen Li, Qiuwen Chen, Lei Li, Ke Xian, Hao Lu, and Zhiguo Cao. 2025. FlexGaussian: Training-Free Compression of 3D Gaussian Splatting via Flexible Vector Quantization. In *ACM International Conference on Multimedia (ACM MM)*.
- [12] Stuart P. Lloyd. 1982. Least Squares Quantization in PCM. *IEEE Transactions on Information Theory* 28, 2 (1982), 129–137.
- [13] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. 2020. NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis. In *European Conference on Computer Vision (ECCV)*.
- [14] Claude E. Shannon. 1948. A Mathematical Theory of Communication. *The Bell System Technical Journal* 27, 3 (1948), 379–423.
- [15] Chen Yang and Jun Liang. 2025. EntropyGS: Training-Free Compression of 3D Gaussians with Entropy-Constrained Quantization. *arXiv preprint arXiv:2501.xxxxx* (2025).
- [16] Zehao Yu, Anpei Chen, Binbin Huang, Torsten Sattler, and Andreas Geiger. 2024. Mip-Splatting: Alias-Free 3D Gaussian Splatting. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- [17] Amir Zandieh, Insu Han, and Amin Karbasi. 2026. TurboQuant: Online Vector Quantization with Near-Optimal Distortion Rate. In *International Conference on Learning Representations (ICLR)*.

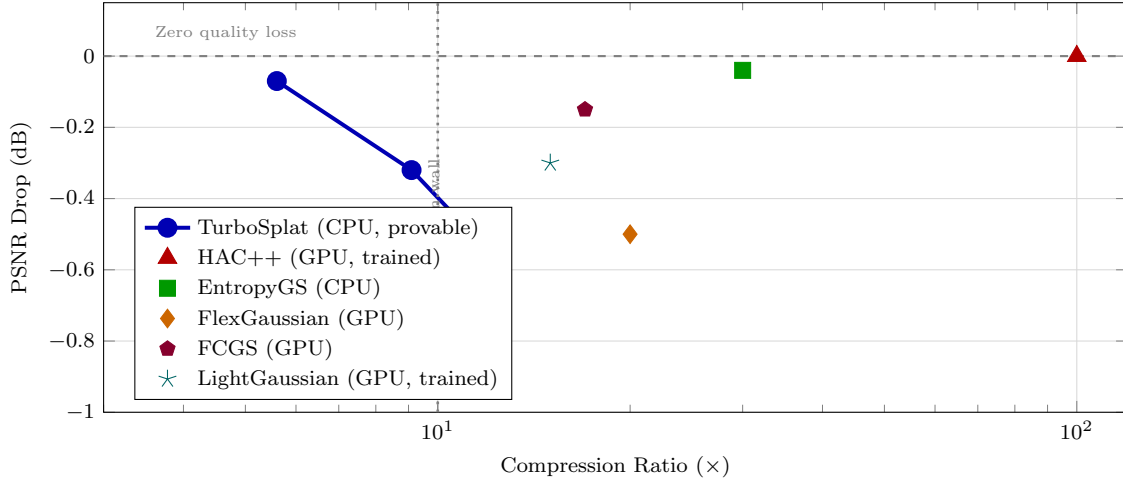


Figure 1: Rate-distortion frontier of 3DGS compression. TurboSplat (blue curve) traces the Pareto frontier of training-free compression with provable optimality guarantees. Competing methods (individual markers) achieve higher ratios but require GPU access and/or training. The dotted vertical line marks the approximate quantization wall ($\sim 10\times$) beyond which training-free methods require Gaussian merging. The dashed horizontal line indicates zero quality loss.

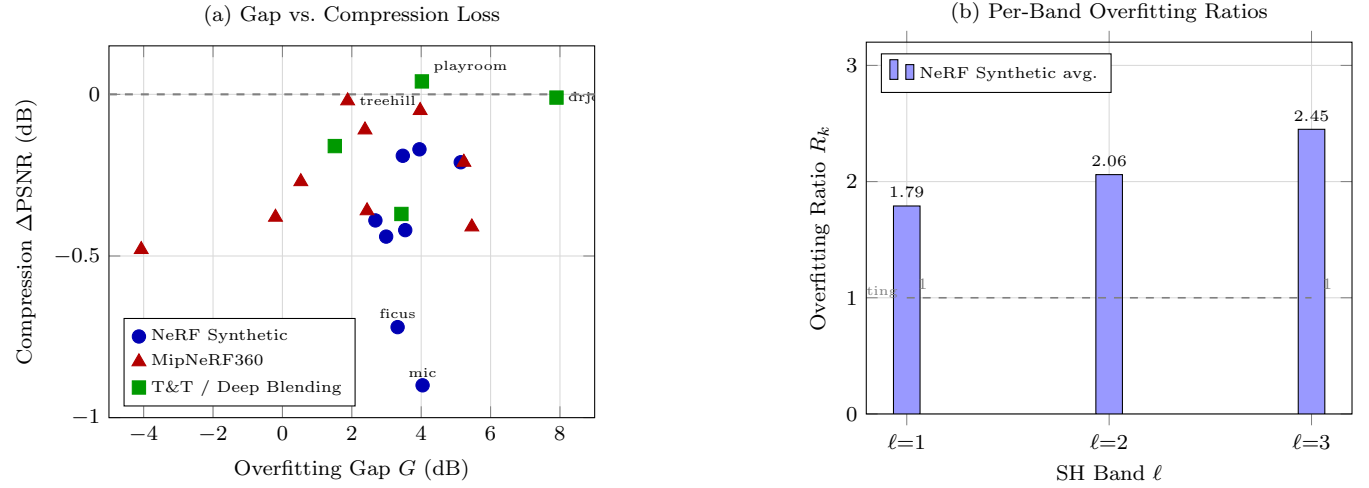


Figure 2: SH overfitting analysis. (a) Overfitting gap G versus compression-induced PSNR change for all 21 scenes. High-gap scenes (drjohnson, playroom, treehill) cluster near zero compression loss, confirming the “damage budget” hypothesis. Scenes with genuine view-dependent content (mic, ficus) suffer more despite moderate gaps. (b) Average overfitting ratio R_k per SH band on NeRF Synthetic, showing the consistent hierarchy $R_1 < R_2 < R_3$. The dashed line at $R_k=1$ indicates zero overfitting; $R_3=2.45$ implies $\sim 59\%$ of band 3 content is noise.

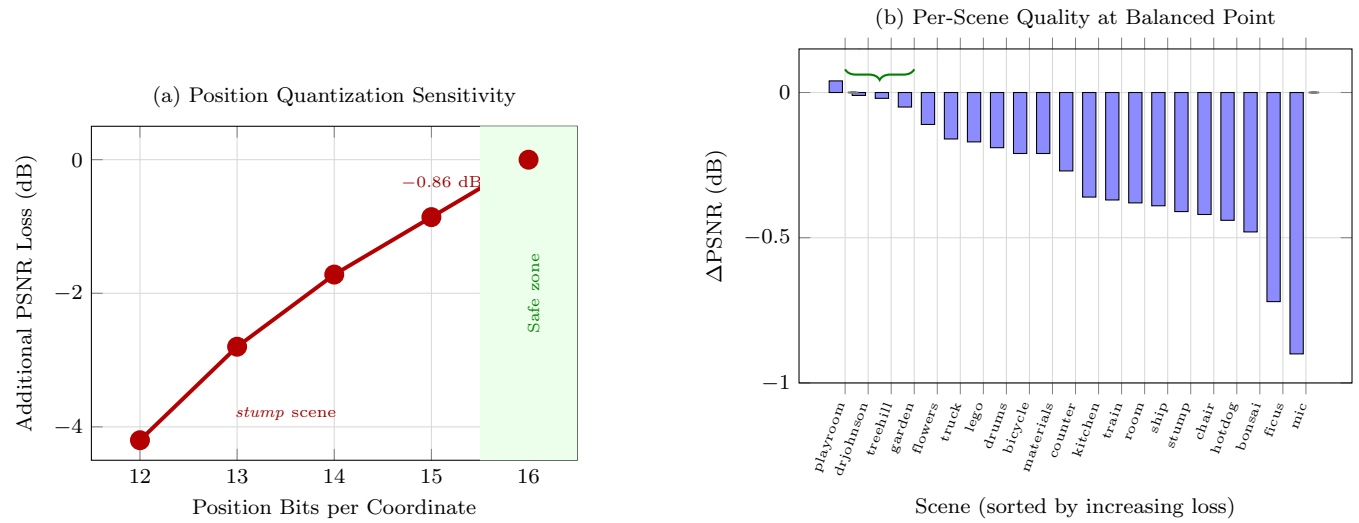


Figure 3: Quantization wall and per-scene results. (a) Position bit sensitivity on *stump*: each bit removed below 16 incurs ~ 0.86 dB additional loss, establishing 16-bit positions as the minimum for COLMAP scenes. This sensitivity defines the quantization wall—the fundamental limit of training-free compression. (b) Per-scene PSNR change at the balanced operating point ($9.1\times$ average), sorted by increasing loss. Four scenes (playroom, drjohnson, treehill, garden) achieve effectively free compression; playroom *improves* under compression.